THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 15	$b = A/l = 360/24$
2	**(d)** 9	$h = 2A/b = 2 \times 72/16$
3	**(d)** 100 cm²	$\frac{1}{2}d(h_1 + h_2) = \frac{1}{2} \times 20 \times (6 + 4)$
4	**(a)** 9720	$18 \times 12 \times 45$
5	**(b)** 20000	$1\ \text{m}^2 = 10000\ \text{cm}^2$
6	**(d)** 8	$a + b = 2A/h = 20$
7	**(d)** 11	$b = A/l = 352/32$
8	**(a)** 13	$h = 2A/b = 2 \times 130/20$
9	**(c)** 144 cm²	$\frac{1}{2}d(h_1 + h_2) = \frac{1}{2} \times 24 \times (5 + 7)$
10	**(a)** 15000	$25 \times 10 \times 60$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$b = A/l = 1200 \div 48 = 25$ m (1); perimeter $= 2(48 + 25) = 146$ m (1).	1+1
12	Area $= 250 \times 80 = 20000$ cm² (1) $= 2$ m² (1).	1+1
13	Area $= \frac{1}{2}d(h_1 + h_2)$ (1) $= \frac{1}{2} \times 26 \times 13 = 169$ cm² (1).	1+1
14	$b = A/l = 1080 \div 54 = 20$ m (1); perimeter $= 2(54 + 20) = 148$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= 32 \times 18 = 576$ m² (1); cost $= 576 \times 12$ (1) $=$ ₹ 6912 (1).	1+1+1
16	$A = \frac{1}{2}(a + b)h$ (1); $130 = \frac{1}{2} \times 26 \times h$ (1); $h = 10$ cm (1).	1+1+1
17	Area $= \frac{1}{2} \times 12 \times 16 = 96$ cm² (2); side $= \sqrt{6^2 + 8^2} = 10$ cm (1).	2+1
	OR: Area $= 45 \times 20 = 900$ m² (1); cost $= 900 \times 8$ (1) $=$ ₹ 7200 (1).	1+1+1
18	$A = \frac{1}{2}(a + b)h$ (1); $144 = \frac{1}{2} \times 36 \times h$ (1); $h = 8$ cm (1).	1+1+1
	OR: Area $= \frac{1}{2} \times 10 \times 24 = 120$ cm² (2); side $= \sqrt{5^2 + 12^2} = 13$ cm (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Trapezium $= \frac{1}{2}(20 + 12) \times 9 = 144$ cm² (2); parallelogram $= 12 \times 9 = 108$ cm² (1); left over $= 36$ cm² (2).	2+1+2
	OR: Area $= 12 \times 9 = 108$ m² (1); length of carpet $= \text{area}/\text{width} = 54$ m (2); cost $= 54 \times 80 =$ ₹ 4320 (2).	1+2+2
20	Trapezium $= \frac{1}{2}(26 + 10) \times 8 = 144$ cm² (2); parallelogram $= 10 \times 8 = 80$ cm² (1); left over $= 64$ cm² (2).	2+1+2
	OR: Area $= 15 \times 8 = 120$ m² (1); length of carpet $= \text{area}/\text{width} = 40$ m (2); cost $= 40 \times 120 =$ ₹ 4800 (2).	1+2+2

Mistakes to avoid:

- Using the slant side instead of the perpendicular height in the area of a parallelogram or triangle.
- Forgetting the factor $\frac{1}{2}$ in the area of a triangle, trapezium or rhombus.
- Mixing units: multiplying a length in metres by one in centimetres; convert first.
- Writing area in cm instead of cm^2, or converting m^2 to cm^2 by multiplying by 100 (it is 10 000).
- Using the sum of the non-parallel sides for a trapezium; only the **parallel** sides enter the formula.
- Adding instead of subtracting when an L-shaped region is obtained by removing a corner.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 30000	$1\ \text{m}^2 = 10000\ \text{cm}^2$
2	**(b)** 9	$a + b = 2A/h = 24$
3	**(a)** 14	$b = A/l = 630/45$
4	**(c)** 7	$h = 2A/b = 2 \times 84/24$
5	**(a)** 108 cm²	$\frac{1}{2}d(h_1 + h_2) = \frac{1}{2} \times 18 \times (8 + 4)$
6	**(c)** 10080	$14 \times 9 \times 80$
7	**(a)** 50000	$1\ \text{m}^2 = 10000\ \text{cm}^2$
8	**(c)** 14	$a + b = 2A/h = 34$
9	**(c)** 17	$b = A/l = 612/36$
10	**(a)** 11	$h = 2A/b = 2 \times 99/18$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area $= 400 \times 150 = 60000$ cm² (1) $= 6$ m² (1).	1+1
12	Area $= \frac{1}{2}d(h_1 + h_2)$ (1) $= \frac{1}{2} \times 34 \times 15 = 255$ cm² (1).	1+1
13	$b = A/l = 770 \div 35 = 22$ m (1); perimeter $= 2(35 + 22) = 114$ m (1).	1+1
14	Area $= 125 \times 160 = 20000$ cm² (1) $= 2$ m² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= 28 \times 25 = 700$ m² (1); cost $= 700 \times 15$ (1) $=$ ₹ 10500 (1).	1+1+1
16	$A = \frac{1}{2}(a + b)h$ (1); $210 = \frac{1}{2} \times 30 \times h$ (1); $h = 14$ cm (1).	1+1+1
17	Area $= \frac{1}{2} \times 18 \times 24 = 216$ cm² (2); side $= \sqrt{9^2 + 12^2} = 15$ cm (1).	2+1
	OR: Area $= 36 \times 24 = 864$ m² (1); cost $= 864 \times 10$ (1) $=$ ₹ 8640 (1).	1+1+1
18	$A = \frac{1}{2}(a + b)h$ (1); $126 = \frac{1}{2} \times 42 \times h$ (1); $h = 6$ cm (1).	1+1+1
	OR: Area $= \frac{1}{2} \times 16 \times 30 = 240$ cm² (2); side $= \sqrt{8^2 + 15^2} = 17$ cm (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Trapezium $= \frac{1}{2}(18 + 8) \times 12 = 156$ cm² (2); parallelogram $= 8 \times 12 = 96$ cm² (1); left over $= 60$ cm² (2).	2+1+2
	OR: Area $= 14 \times 10 = 140$ m² (1); length of carpet $= \text{area}/\text{width} = 35$ m (2); cost $= 35 \times 90 =$ ₹ 3150 (2).	1+2+2
20	Trapezium $= \frac{1}{2}(30 + 14) \times 6 = 132$ cm² (2); parallelogram $= 14 \times 6 = 84$ cm² (1); left over $= 48$ cm² (2).	2+1+2
	OR: Area $= 18 \times 12 = 216$ m² (1); length of carpet $= \text{area}/\text{width} = 72$ m (2); cost $= 72 \times 75 =$ ₹ 5400 (2).	1+2+2

Mistakes to avoid:

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 150 cm²	$\frac{1}{2}d(h_1+h_2)=\frac{1}{2}\times 30\times(7+3)$
2	**(a)** 15750	$30\times 15\times 35$
3	**(c)** 70000	$1\ \text{m}^2=10000\ \text{cm}^2$
4	**(c)** 7	$a+b=2A/h=18$
5	**(d)** 19	$b=A/l=532/28$
6	**(b)** 8	$h=2A/b=2\times 120/30$
7	**(a)** 112 cm²	$\frac{1}{2}d(h_1+h_2)=\frac{1}{2}\times 16\times(9+5)$
8	**(b)** 13200	$16\times 11\times 75$
9	**(d)** 120000	$1\ \text{m}^2=10000\ \text{cm}^2$
10	**(d)** 10	$a+b=2A/h=28$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area $=\frac{1}{2}d(h_1+h_2)$ (1) $=\frac{1}{2}\times 22\times 17=187$ cm² (1).	1+1
12	$b=A/l=960\div 64=15$ m (1); perimeter $=2(64+15)=158$ m (1).	1+1
13	Area $=300\times 200=60000$ cm² (1) $=6$ m² (1).	1+1
14	Area $=\frac{1}{2}d(h_1+h_2)$ (1) $=\frac{1}{2}\times 40\times 15=300$ cm² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $=50\times 16=800$ m² (1); cost $=800\times 9$ (1) $=$ ₹ 7200 (1).	1+1+1
16	$A=\frac{1}{2}(a+b)h$ (1); $300=\frac{1}{2}\times 50\times h$ (1); $h=12$ cm (1).	1+1+1
17	Area $=\frac{1}{2}\times 24\times 32=384$ cm² (2); side $=\sqrt{12^2+16^2}=20$ cm (1).	2+1
	OR: Area $=42\times 15=630$ m² (1); cost $=630\times 14$ (1) $=$ ₹ 8820 (1).	1+1+1
18	$A=\frac{1}{2}(a+b)h$ (1); $224=\frac{1}{2}\times 28\times h$ (1); $h=16$ cm (1).	1+1+1
	OR: Area $=\frac{1}{2}\times 14\times 48=336$ cm² (2); side $=\sqrt{7^2+24^2}=25$ cm (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Trapezium $=\frac{1}{2}(24+16)\times 11=220$ cm² (2); parallelogram $=16\times 11=176$ cm² (1); left over $=44$ cm² (2).	2+1+2
	OR: Area $=16\times 15=240$ m² (1); length of carpet $=\text{area}/\text{width}=40$ m (2); cost $=40\times 60=$ ₹ 2400 (2).	1+2+2
20	Trapezium $=\frac{1}{2}(22+6)\times 10=140$ cm² (2); parallelogram $=6\times 10=60$ cm² (1); left over $=80$ cm² (2).	2+1+2
	OR: Area $=20\times 9=180$ m² (1); length of carpet $=\text{area}/\text{width}=36$ m (2); cost $=36\times 150=$ ₹ 5400 (2).	1+2+2

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